

Chapter 6. Biomacromolecules of Life – Nucleic Acid

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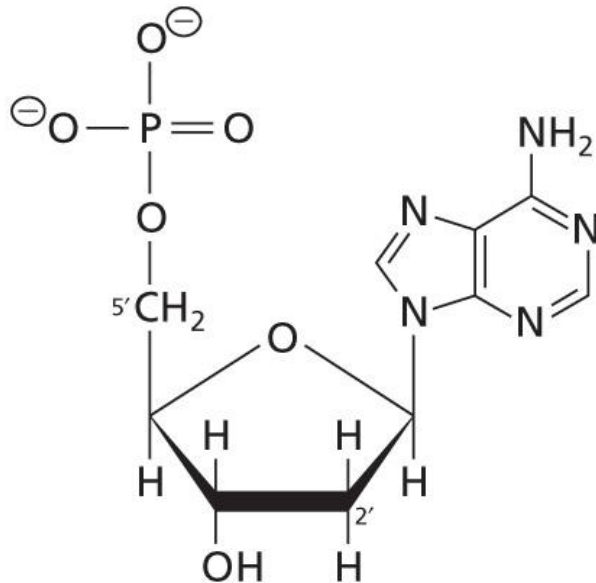
1. Nucleotides
2. Double helix
3. Nucleosome and chromatin
4. RNA structures

Reference:

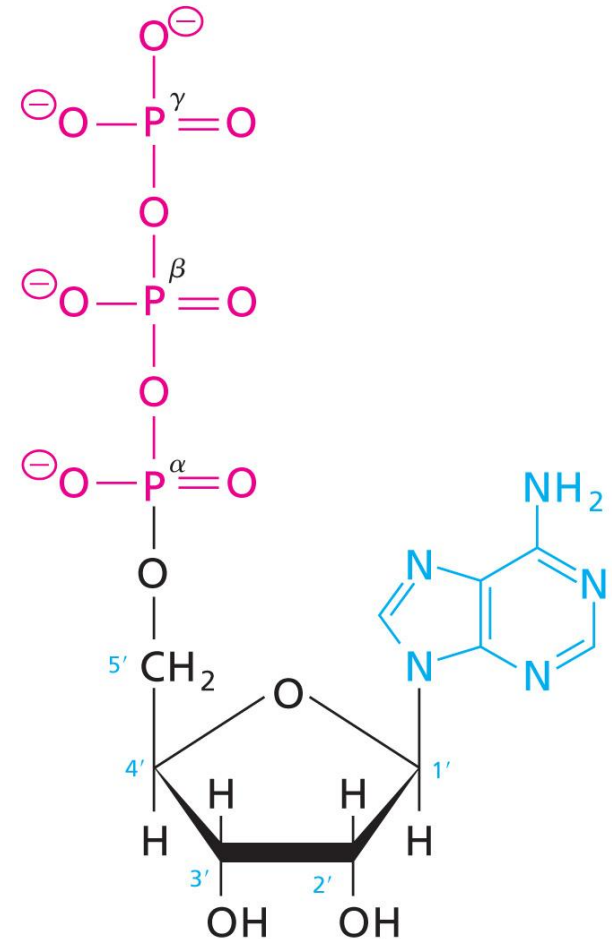
Principles of Biochemistry, by Horton et al.

1. Nucleotides

Nucleotides are the building blocks of biopolymers – nucleic acids



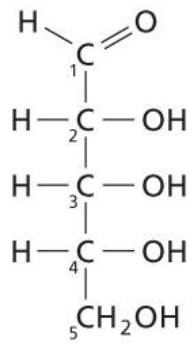
2'-Deoxyadenosine 5'-monophosphate
(Deoxyadenylate, dAMP)



Adenosine triphosphate
(ATP)

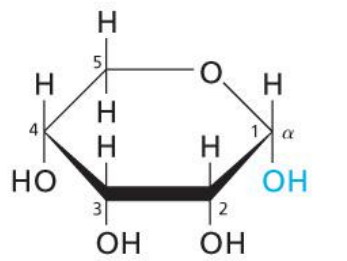
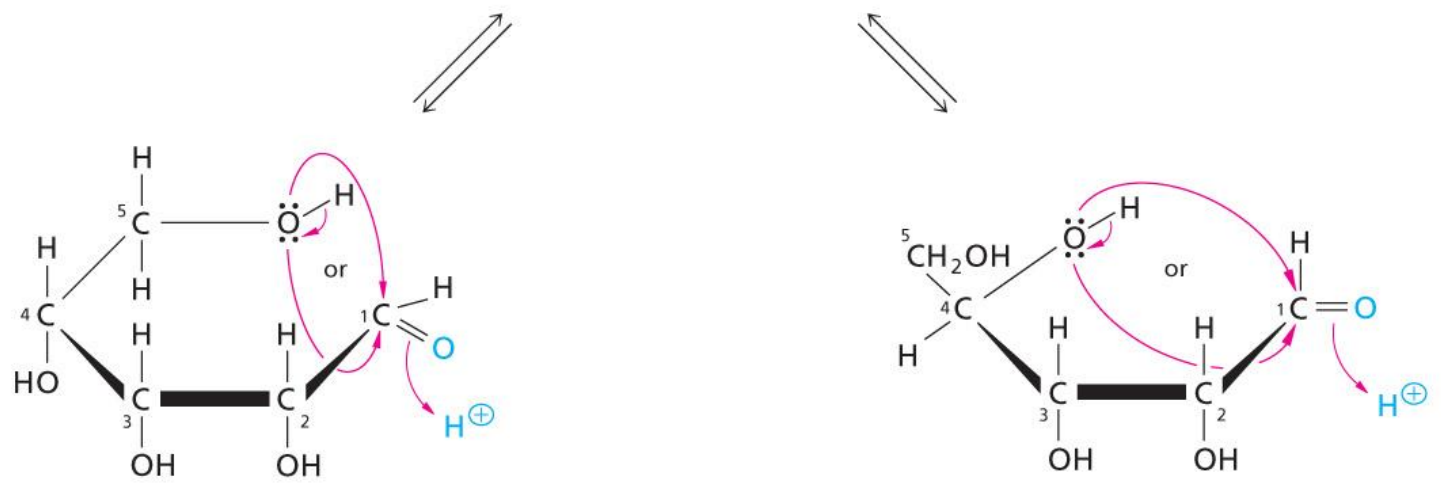
1. Nucleotides

D-Ribose structures

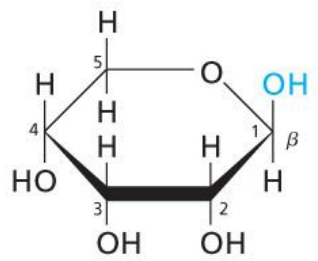


Note: The absolute configurations remain the same, although the designation of R/S configurations may change.

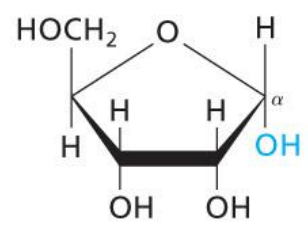
D-Ribose



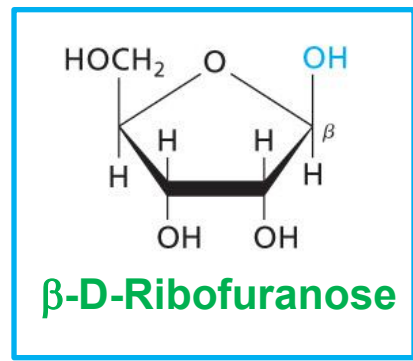
α -D-Ribopyranose



β -D-Ribopyranose



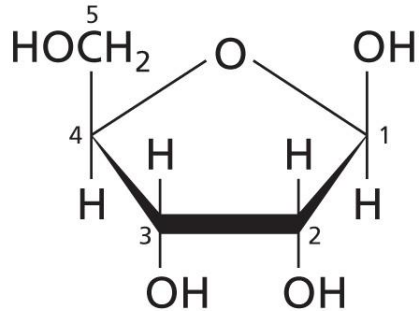
α -D-Ribofuranose



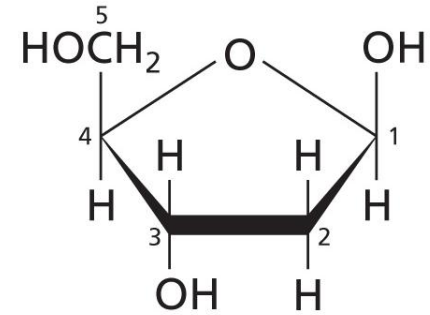
β -D-Ribofuranose

1. Nucleotides

Ribose and deoxyribose structures:

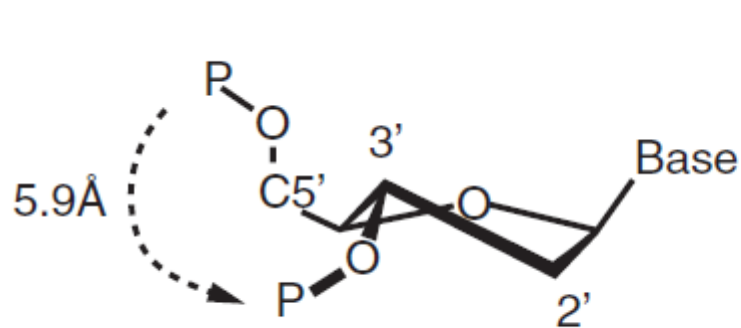


Ribose
(β -D-Ribofuranose)



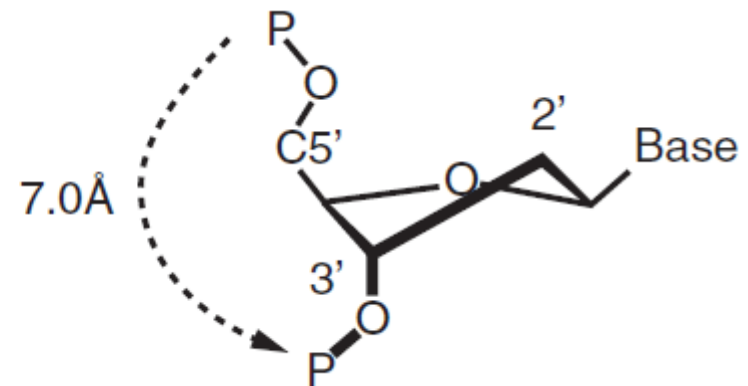
Deoxyribose
(2-Deoxy- β -D-ribofuranose)

Two types of sugar pucker most commonly found in nucleic acids:



C3'-endo

Prevalent in A-form DNA and RNA duplexes



C2'-endo

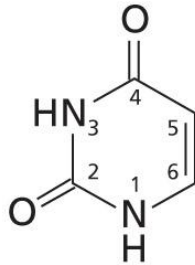
Characteristic of B-form DNA duplexes 4

1. Nucleotides

Base structures: The two types of heterocyclic bases are derivatives of **purine** and **pyrimidine**

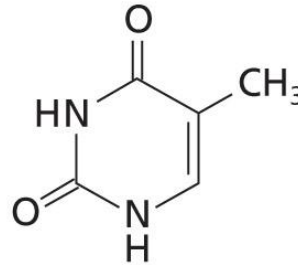


Pyrimidine



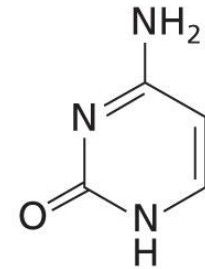
Uracil

(2,4-Dioxypyrimidine)



Thymine

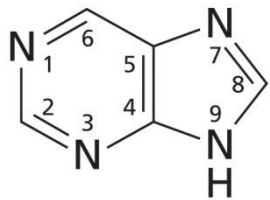
(2,4-Dioxo-5-methylpyrimidine)



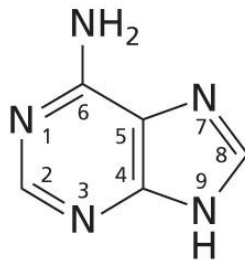
Cytosine

(2-Oxo-4-aminopyrimidine)

PYRIMIDINES

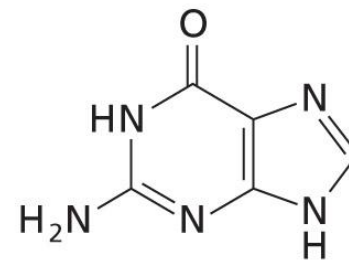


Purine



Adenine

(6-Aminopurine)



Guanine

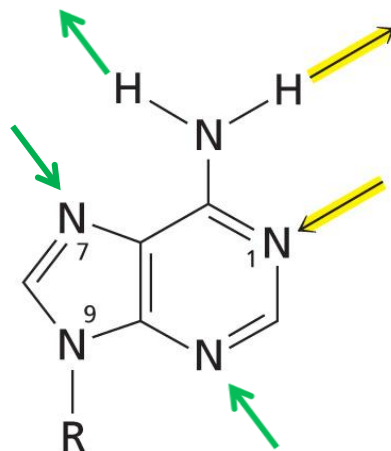
(2-Amino-6-oxopurine)

PURINES

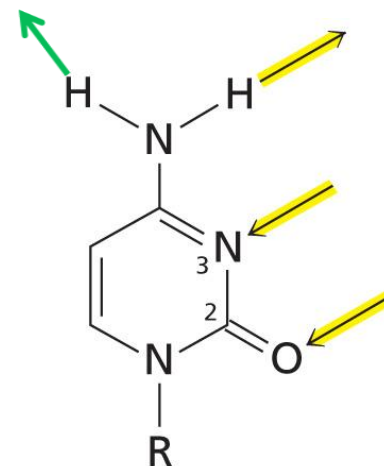
1. Nucleotides

Base structures: Hydrogen bond donors and acceptors

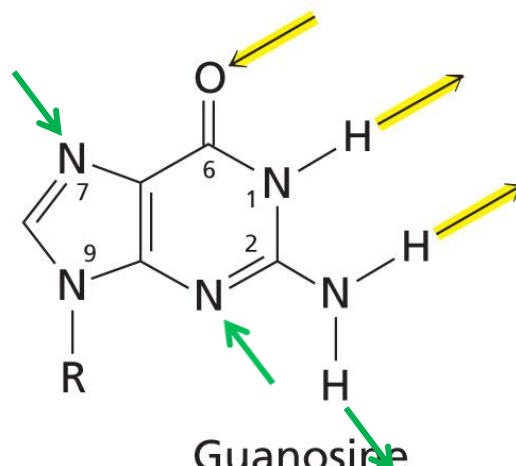
An exocyclic amine group can't be a **hydrogen bond acceptor**.



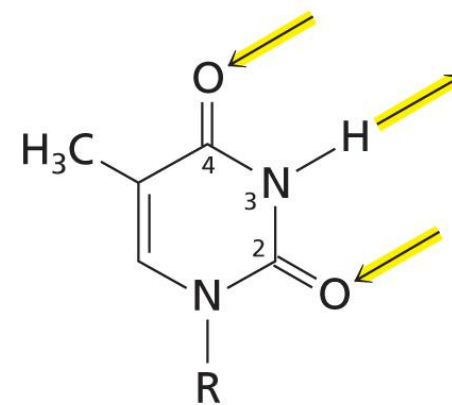
Adenosine



Cytidine



Guanosine



Thymidine

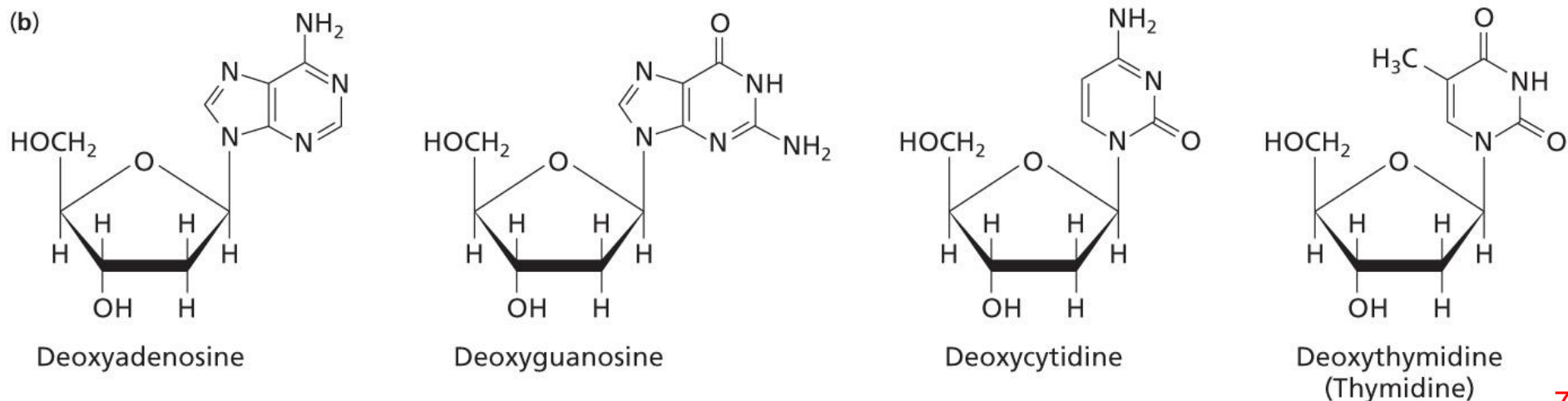
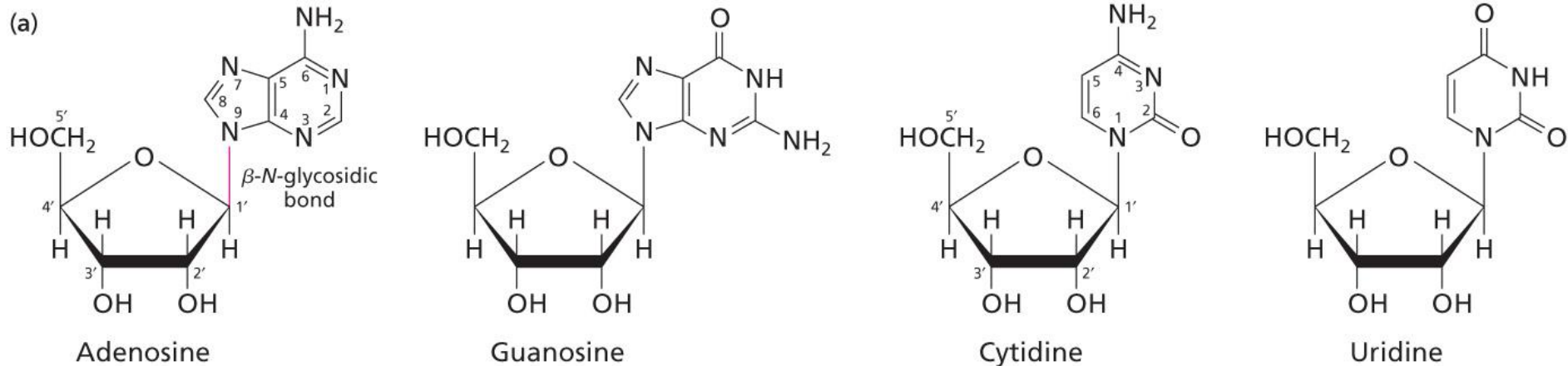
Online link:

<http://www.chemguide.co.uk/organicprops/aniline/background.html>

1. Nucleotides

Nucleoside structure: A nitrogenous base with a ribose

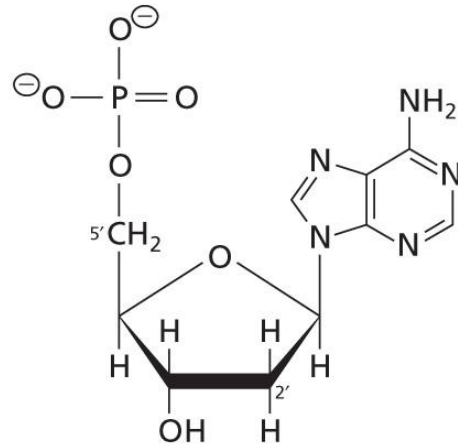
Ribonucleoside and **deoxyribonucleoside**



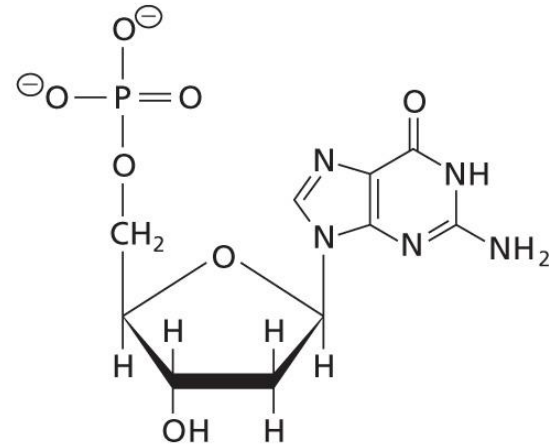
1. Nucleotides

Nucleotide structure: A nitrogenous base, a ribose and a phosphate

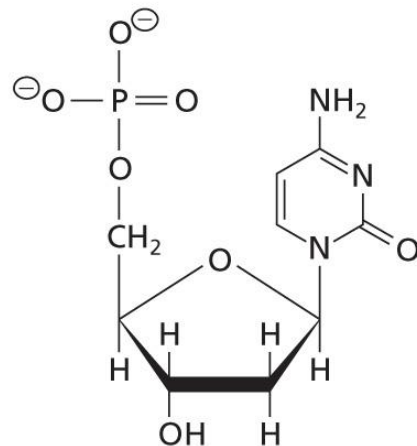
Shown below are deoxyribonucleotide structures:



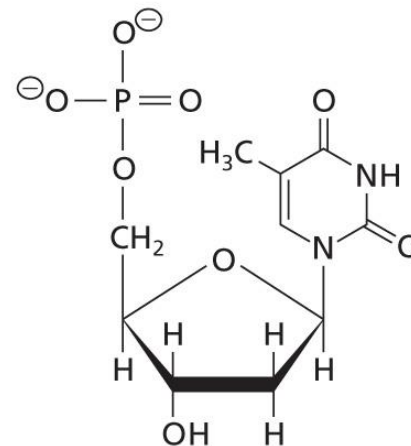
2'-Deoxyadenosine 5'-monophosphate
(Deoxyadenylate, dAMP)



2'-Deoxyguanosine 5'-monophosphate
(Deoxyguanylate, dGMP)



2'-Deoxycytidine 5'-monophosphate
(Deoxycytidylate, dCMP)



2'-Deoxythymidine 5'-monophosphate
(Thymidylate, dTMP)

1. Nucleotides

Nomenclature of **bases**, **nucleosides**, and **nucleotides**

Base	Ribonucleoside	Ribonucleotide (5'-monophosphate)
Adenine (A)	Adenosine	Adenosine 5'-monophosphate (AMP); adenylate ^a
Guanine (G)	Guanosine	Guanosine 5'-monophosphate (GMP); guanylate ^a
Cytosine (C)	Cytidine	Cytidine 5'-monophosphate (CMP); cytidylate ^a
Uracil (U)	Uridine	Uridine 5'-monophosphate (UMP); uridylate ^a
Base	Deoxyribonucleoside	Deoxyribonucleotide (5'-monophosphate)
Adenine (A)	Deoxyadenosine	Deoxyadenosine 5'-monophosphate (dAMP); deoxyadenylate ^a
Guanine (G)	Deoxyguanosine	Deoxyguanosine 5'-monophosphate (dGMP); deoxyguanylate ^a
Cytosine (C)	Deoxycytidine	Deoxycytidine 5'-monophosphate (dCMP); deoxycytidylate ^a
Thymine (T)	Deoxythymidine or thymidine	Deoxythymidine 5'-monophosphate (dTMP); deoxythymidylate ^a or thymidylate ^a

^aAnionic forms of phosphate esters predominant at pH 7.4.

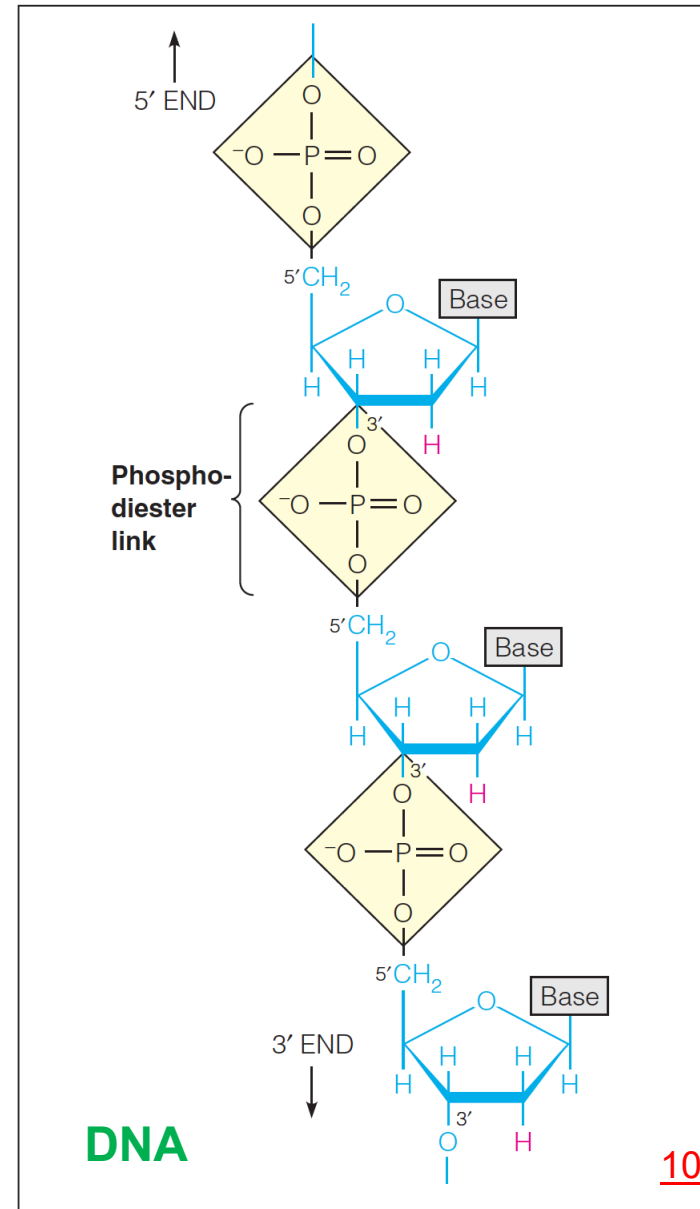
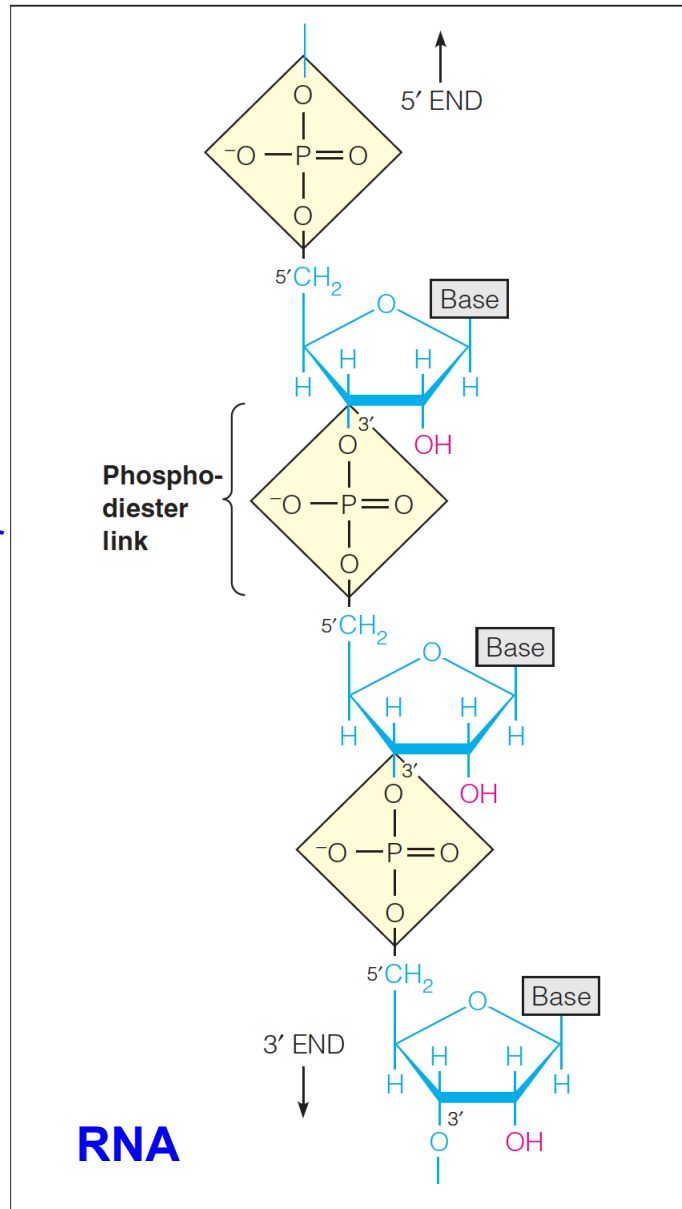
2. Double helix

RNA contains the sugar **ribose**, **DNA** has the **deoxyribose**

Directionality:
5' and 3'

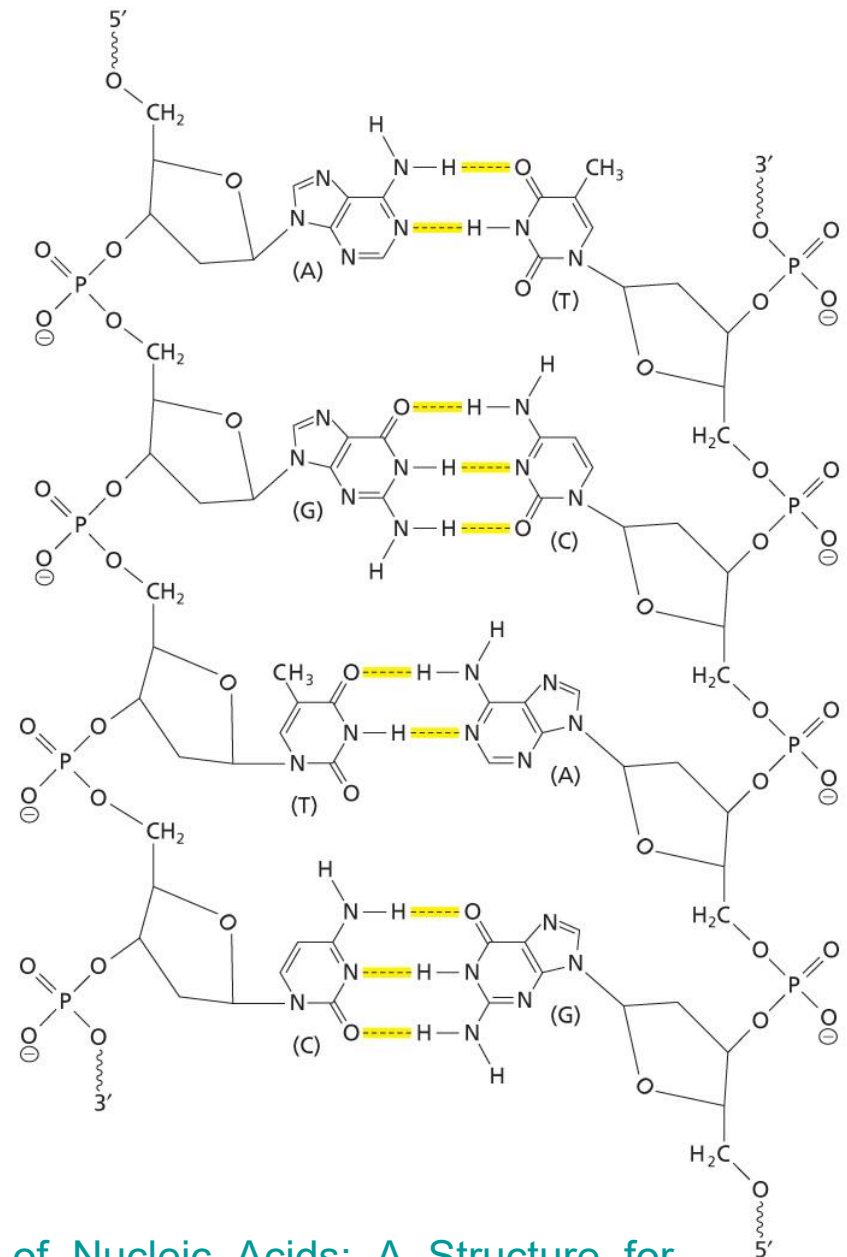
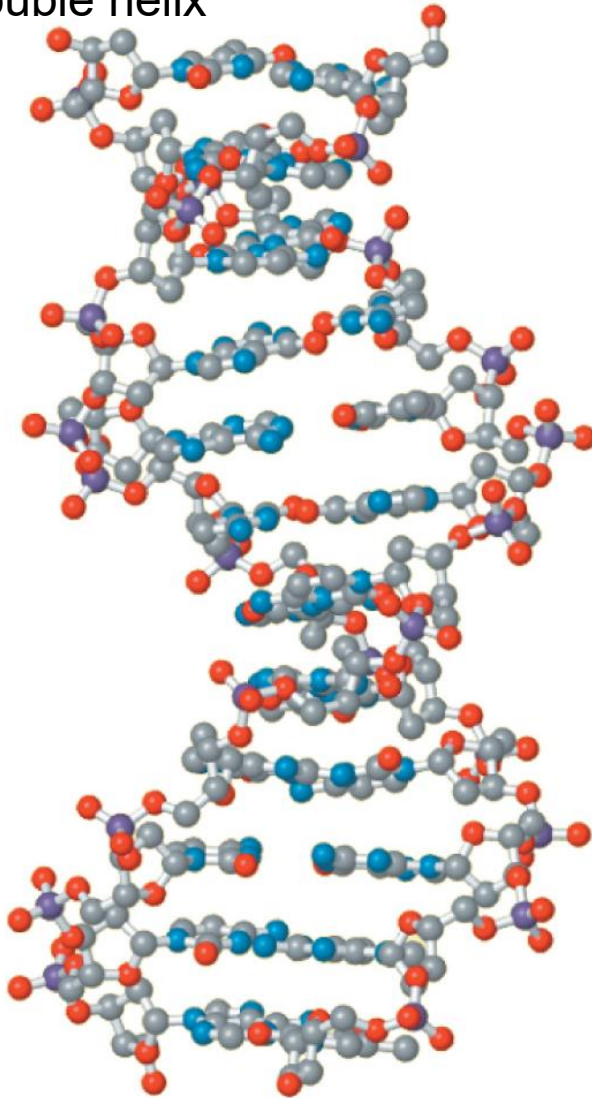
Primary structure:
The sequence

Nucleic acids can
be linear or circular



2. Double helix

DNA double helix

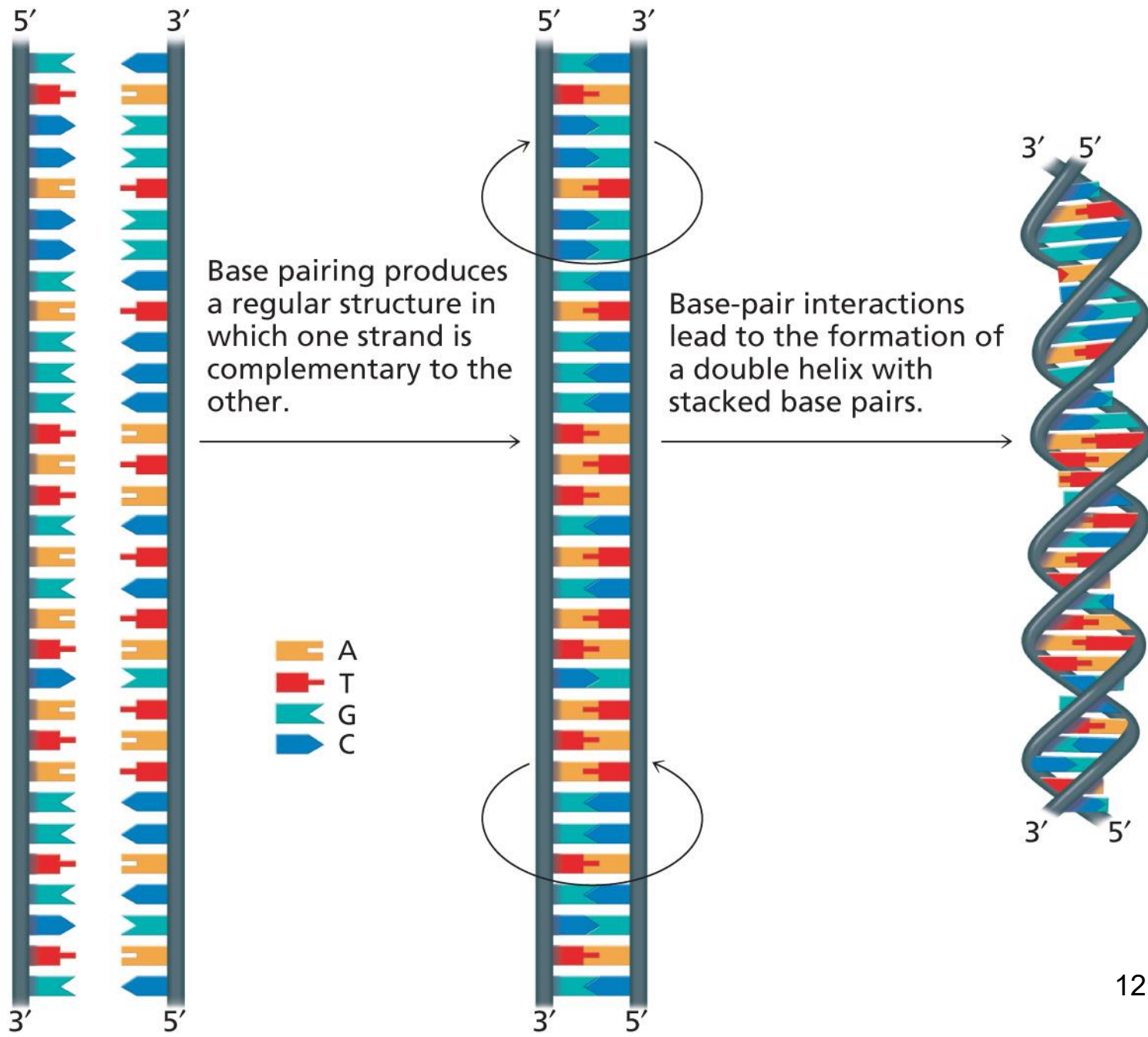


Online links:

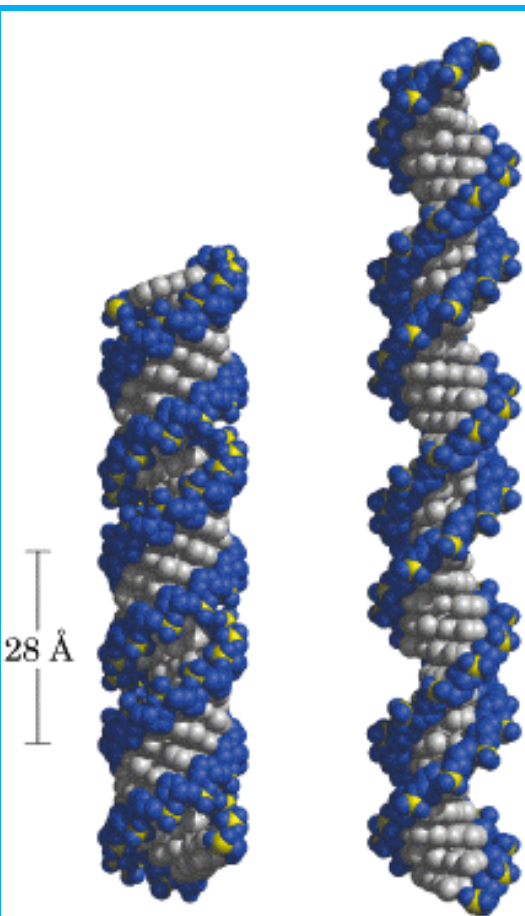
http://en.wikipedia.org/wiki/Molecular_Structure_of_Nucleic_Acids:_A_Structure_for_Deoxyribose_Nucleic_Acid

<https://www.nature.com/articles/d41586-023-01313-5>

2. Double helix



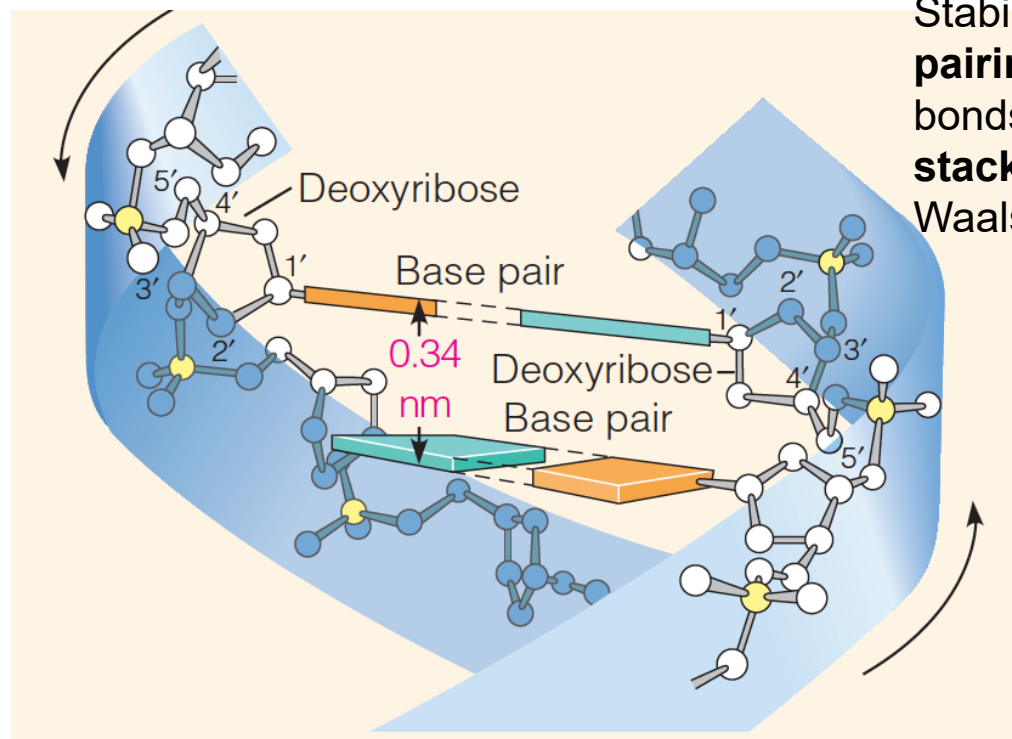
2. Double helix



A form

B form

Both structures have
33 base pairs



Stabilized by **base pairing** (hydrogen bonds) and **base stacking** (van der Waals interaction).

	A Form	B Form
Direction of helix rotation	Right	Right
Number of residues per turn (n)	11	10
Rotation per residue ($= 360^\circ/n$)	33°	36°
Rise ^a in helix per residue (h)	0.255 nm	0.34 nm
Pitch ^a of helix ($= nh$)	2.8 nm	3.4 nm

3. Nucleosome and chromatin

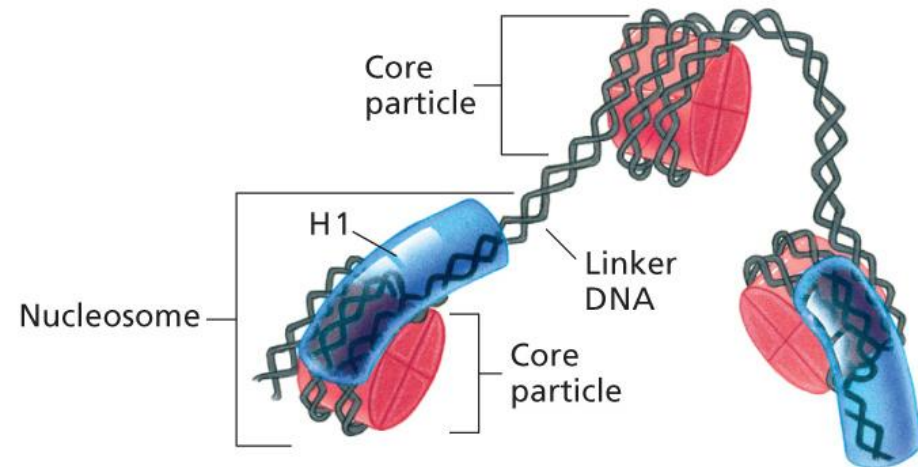
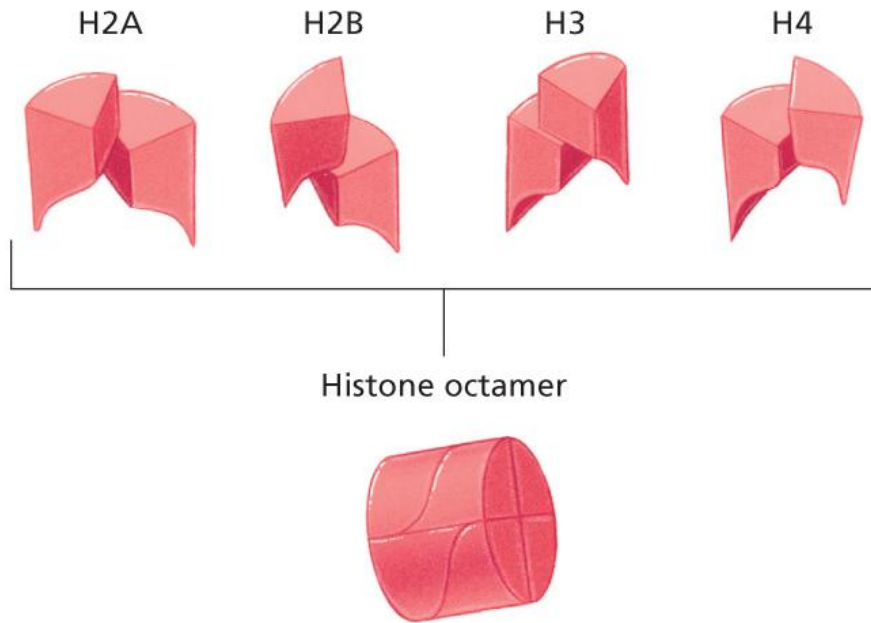
Histone octamer: Two copies of H2A, H2B, H3, and H4

Nucleosome core particle: Histone octamer and 146 bp of DNA

Nucleosome: A nucleosome core particle plus histone H1 and linker DNA (about 54 bp)

Histone H1 binds to the core particle and to linker DNA

On the right is an **extended chromatin** showing the “beads-on-a-string” organization



Online link:

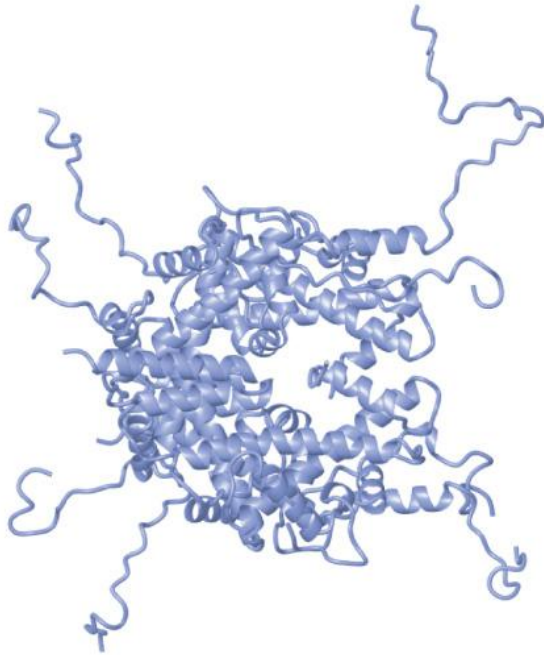
<http://en.wikipedia.org/wiki/Nucleosome>

3. Nucleosome and chromatin

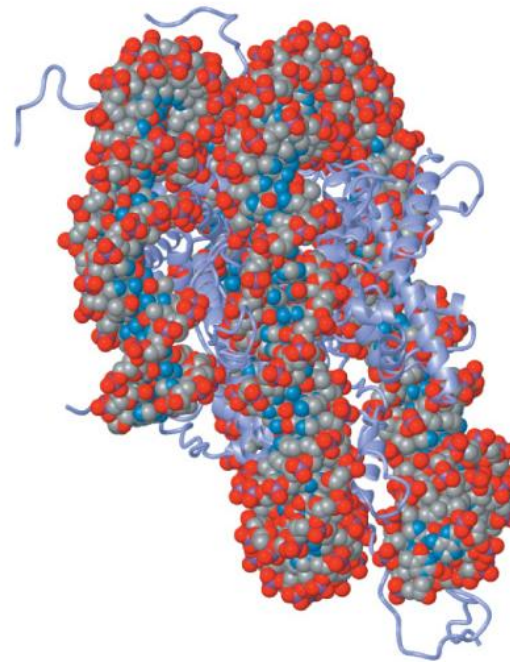
Structure of nucleosome core particle

The histone octamer is shaped like a flat disk (with a width of about 5 nm), with positively charged grooves (**amine groups**) to accommodate the **sugar-phosphate** backbone of DNA

DNA (about 146 bp) wraps around the histone octamer. Formation of nucleosome core particle reduces the **DNA end-to-end distance**.



Histone octamer



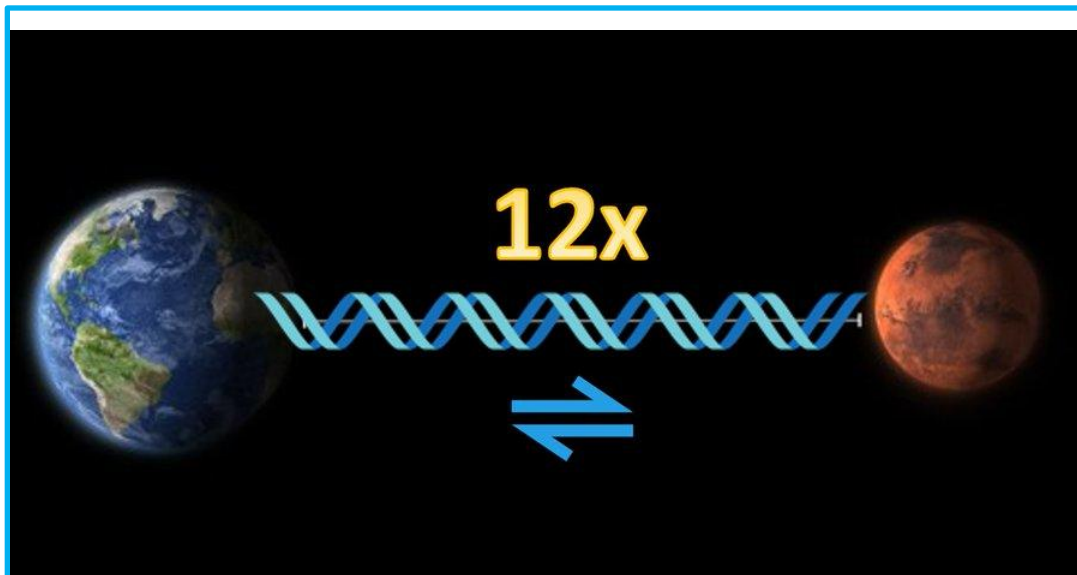
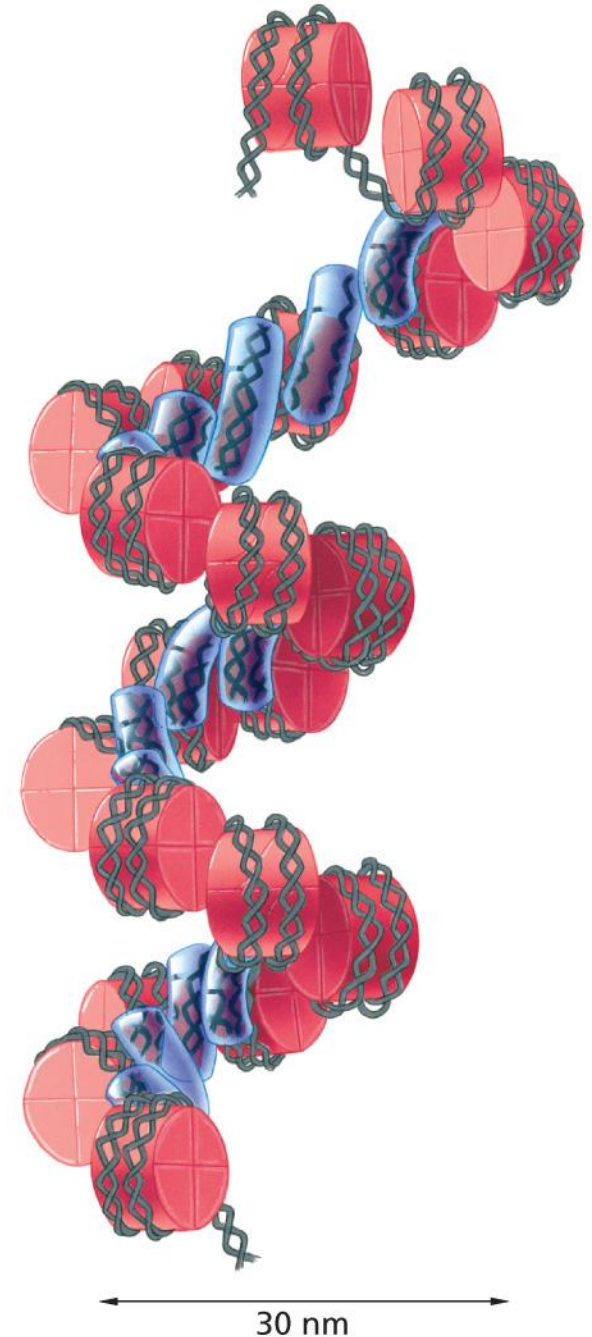
Nucleosome core particle:
Histone octamer bound to DNA

3. Nucleosome and chromatin

A model of the 30-nm chromatin fiber.

The nucleosomes associate through contacts between adjacent histone H1 molecules

The 30-nm fiber forms a left-handed helix and achieves a further 4-fold reduction in chromosome **end-to-end distance**



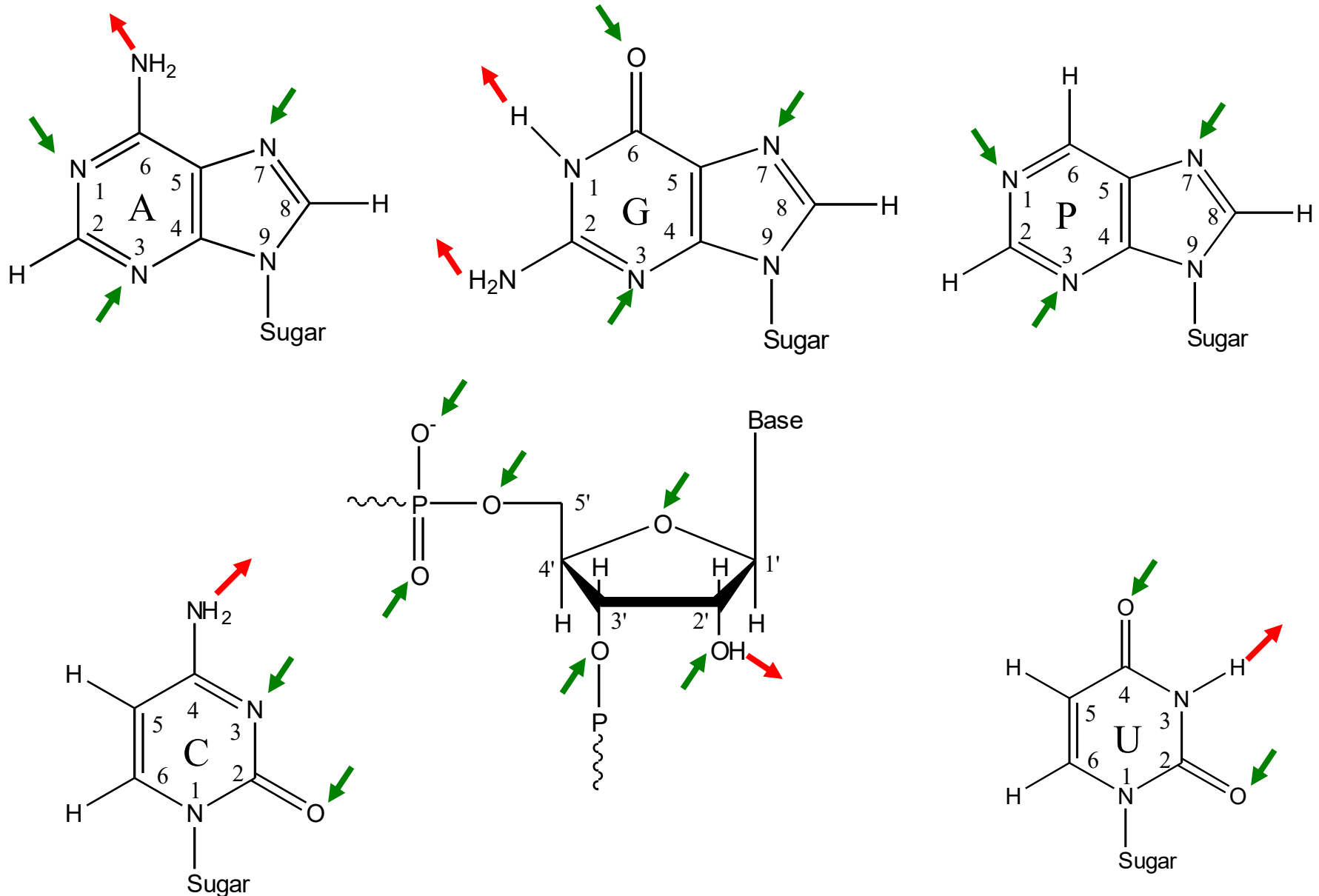
You have about 10 trillion cells in your body, so if you stretched the DNA in all the cells out, end to end, they'd stretch over 750 million miles. Mars is only about 34 million miles away, so all your DNA would stretch to Mars and back almost 12 times.

3. Nucleosome and chromatin

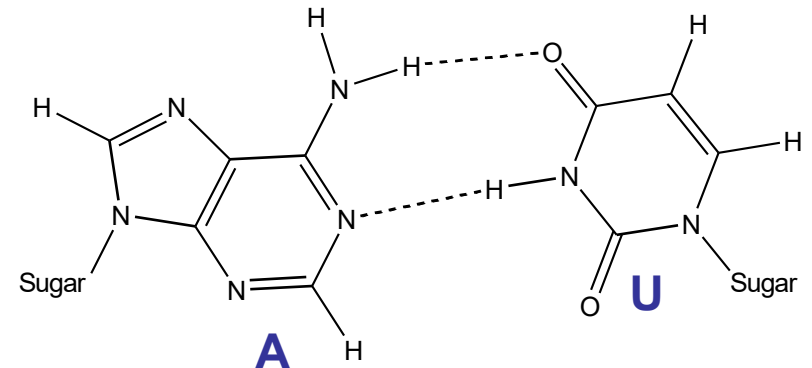
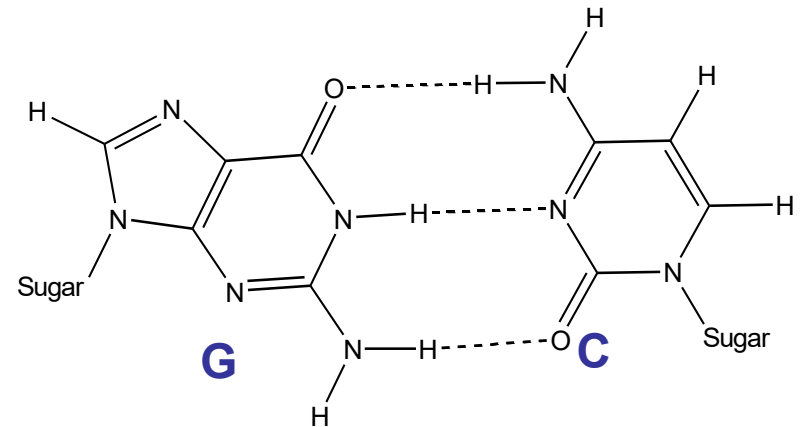
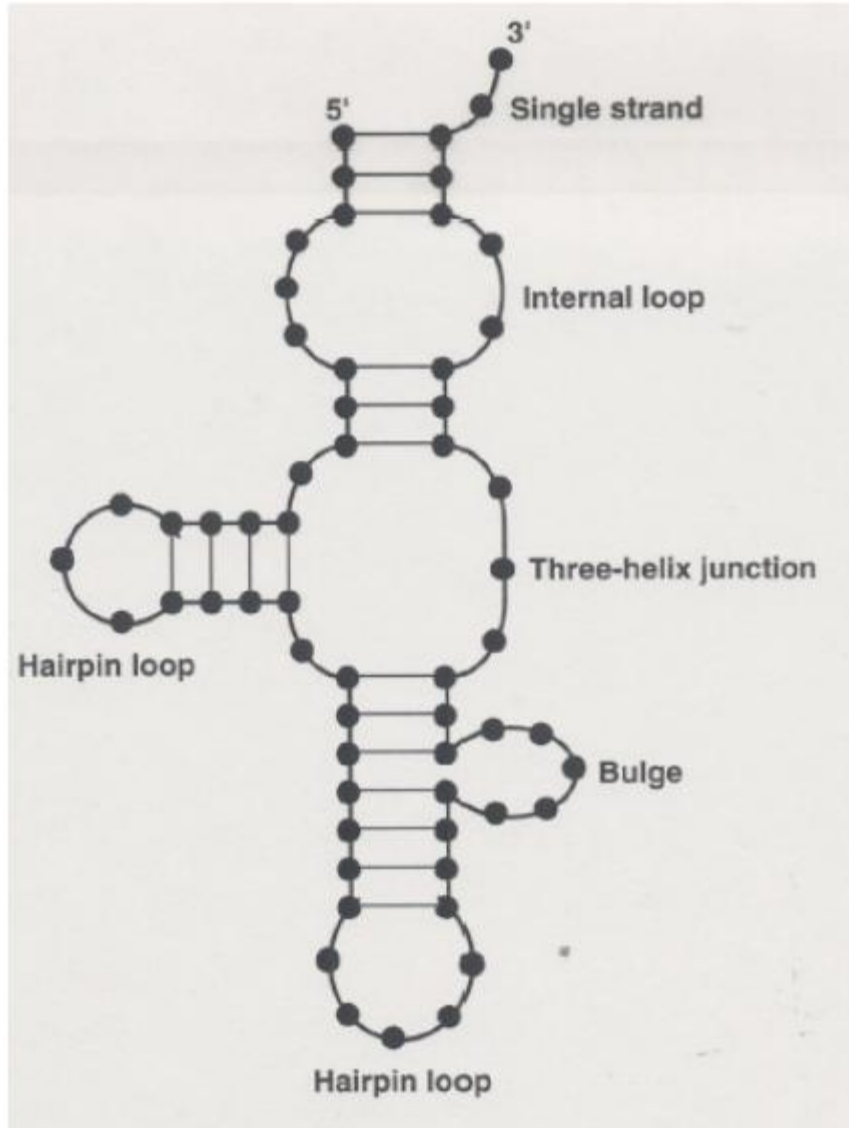


4. RNA secondary structures

Base and nucleotide structures: Hydrogen bond **donors** and **acceptors**



4. RNA secondary structures



The straight short lines stand for Watson-Crick base pairs (G-C and A-U).

4. RNA secondary structures



Single strand



A-form helix



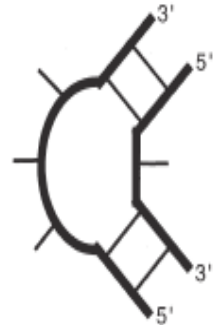
Dangling end



1 × 1 internal loop



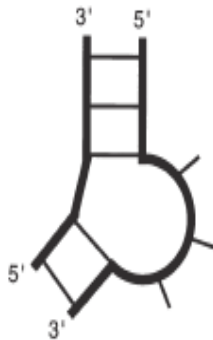
Symmetric (3 × 3) internal loop



Asymmetric (1 × 3) internal loop



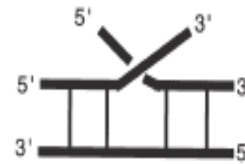
Single nucleotide bulge



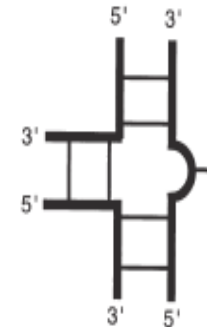
3-nt bulge



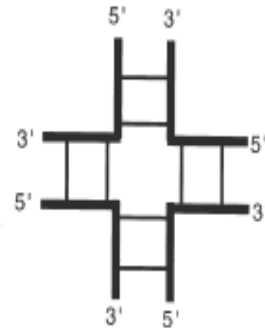
Hairpin loop



Two-stem junction or coaxial stack



Three-stem junction



Four-stem junction

The straight short lines stand for Watson-Crick base pairs (G-C and A-U).

4. RNA structures

Self-complementarity in a base sequence allows a chain to fold back on itself and form an RNA secondary structure containing **stems** (with Watson-Crick G-C and A-U pairs), and **hairpins** and **junctions**...

Shown below are the secondary structure and tertiary structure of a **tRNA**

